DO NOW — RECALL

LMC branching and debugging

Name: _____ Class: _____ Date: _____

1. What does STA do?

2. What does LMC HLT do?

3. Why must a loop's counter change?

4. Full adder: inputs A=1, B=0, carry-in=1. Find sum and carry-out.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Copies ACC to mailbox.

2. Model answer

Stops execution.

3. Model answer

Without updating a tested state, the loop may not terminate.

4. Model answer

$1+0+1=10_2$; sum=0, carry-out=1.

Vocabulary — know these terms

BRA — Branch Always

BRZ — Branch if Zero

BRP — Branch if non-negative

mailbox — LMC memory cell

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Check next state

BRZ not taken

OUT prints 4

HLT stops execution.

IMPROVE: Explain one correction to your own reasoning.

NEW LEARNING

Read, annotate and explain

Understand

BRP tests whether ACC is non-negative; BRZ tests zero; BRA is unconditional.

Trace

Load the counter, output it and subtract one. Test BRP immediately: save the new value and loop only when it is non-negative.

Explain

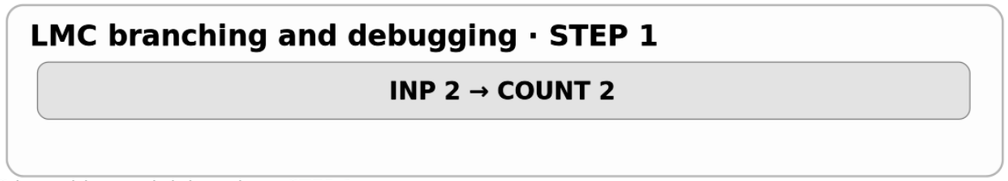
After outputting 0, SUB ONE underflows; BRP is not taken, so HLT runs without storing a negative mailbox value.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

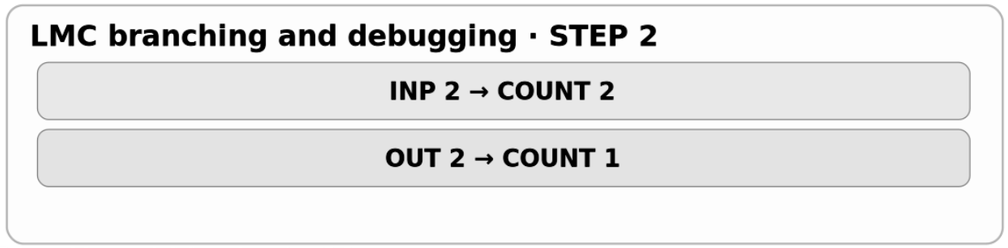
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



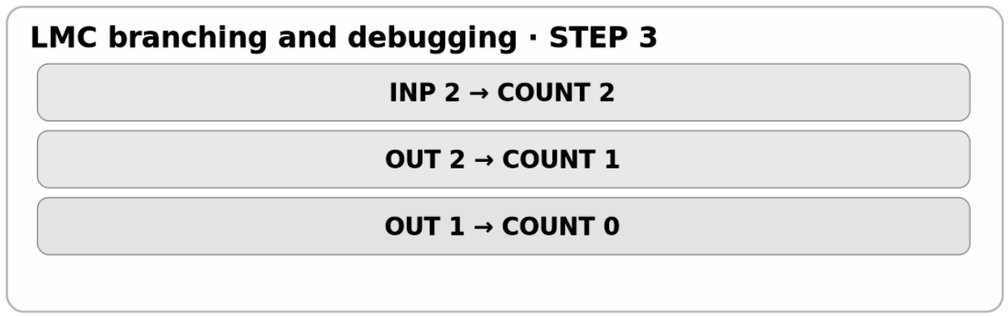
LMC branching and debugging · STEP 1

New Learning



LMC branching and debugging · STEP 2

New Learning



LMC branching and debugging · STEP 3

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

An unconditional backward branch always terminates.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

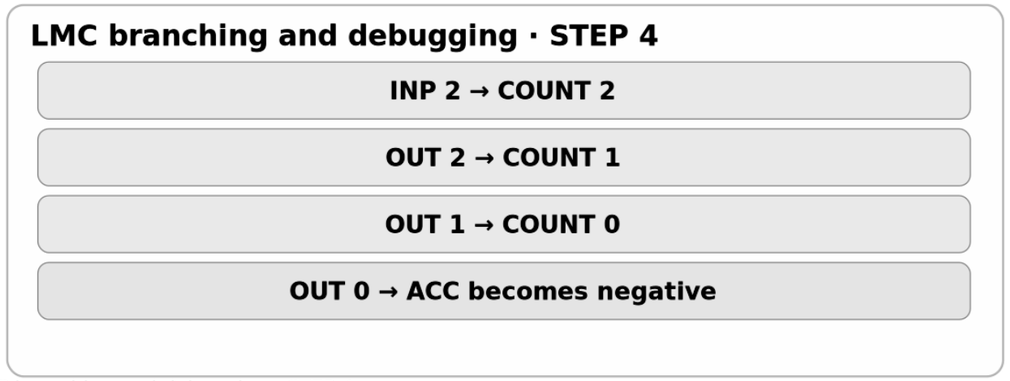
Question 3

Explain why an unconditional backward branch without a tested exit can run forever, and provide a safe counter-based solution. [6]

New Learning · original lesson diagrams

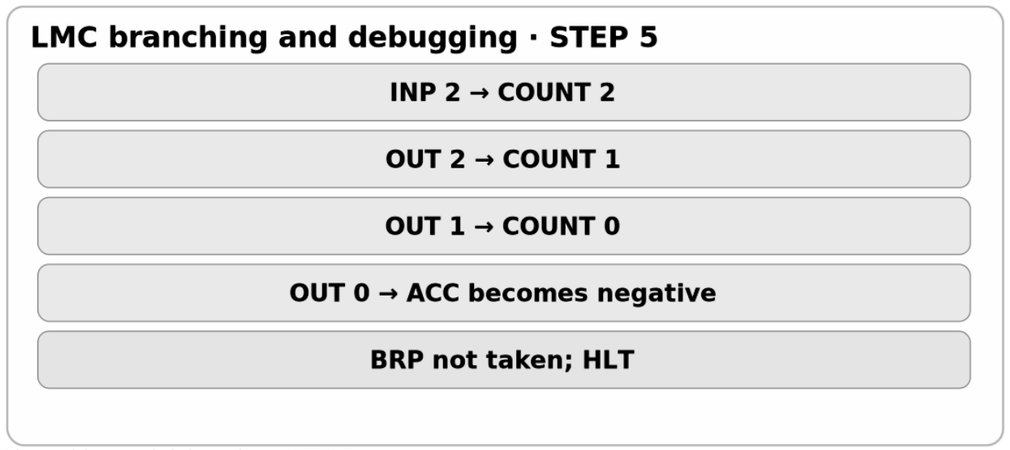
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



LMC branching and debugging · STEP 4

New Learning



LMC branching and debugging · STEP 5

I DO — WORKED EXAMPLE

Follow the worked method

Trace BRZ when input is 0.

Predict first

INP sets ACC to 0

Check next state

BRZ ZERO takes branch

Check next state

LDA ONE loads 1

Check next state

OUT displays 1.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define BRA, BRZ and BRP and explain their effect on PC. [4]

Question 2

Input 3: trace INP; BRZ STOP; SUB ONE; OUT; HLT; ONE DAT 1; STOP HLT. [6]

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first INP makes ACC 4

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

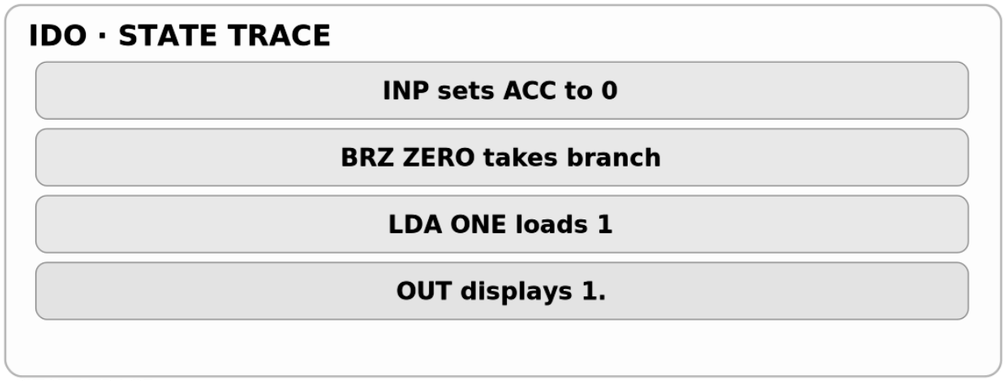
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace BRZ when input is 0.

INP sets ACC to 0



IDO · STATE TRACE

WE DO — GUIDED PRACTICE

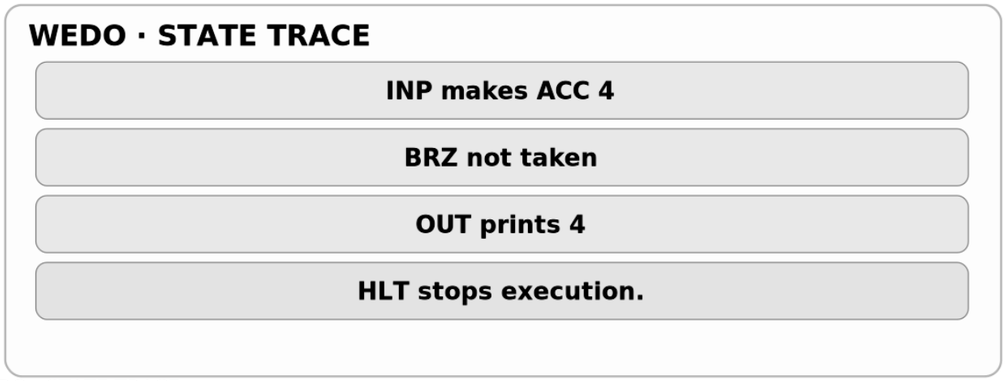
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace the same code with input 4.

INP makes ACC 4



WEDO · STATE TRACE